

This assignment will not be graded and is only for practice.

A relation is said to be in the first normal form (1NF) when,

2 points

- ☐ it does not contain any partial dependency
- ☐ it does not have multivalued attributes
- ☐ it has multivalued attributes
- ☐ All the above

Which of the following normal forms exist(s)?

2 points

- ☐ ETNF
- ☐ BCNF
- ☐ EKNF
- ☐ All the above

Consider the relation **Customer** as shown in Table 1.

2 points

Table 1: **Customer**

<i>CId</i>	<i>Name</i>	<u><i>Contact</i></u>
S001	David	11111
S002	Warner	12233
S002	Warner	12222
S003	Mathew	23221
S003	Mathew	23223
S005	Hayden	22111

The relation **Customer**

- ☐ is in 2NF
- ☐ is in 1NF

- ☐ is neither in 1NF nor in 2NF
- ☐ has multivalued attributes

Consider the following statements:

2 points

1. A deletion anomaly occurs when you delete a record that should not be deleted.
2. An insertion anomaly occurs when you insert inconsistent or unrelated information into a table.
3. Normalization or Schema Refinement is a technique of organizing the data in the database.

- ☐ Only Statements 1 & 2 are correct.
- ☐ Only Statement 3 is correct.
- ☐ Only Statement 2 is correct.
- ☐ All the statements are correct.

A relation **R** is said to be in 2NF when

2 points

- ☐ **R** is not in BCNF
- ☐ **R** is in 1NF and does not contain any partial dependencies
- ☐ **R** does not contain any transitive dependency
- ☐ All the above

What is/are the main purpose(s) of normalization?

2 points

- ☐ To eliminate functional dependencies
- ☐ To eliminate redundant data
- ☐ To reduce anomalies in a database
- ☐ None of the above

Consider the relation **Customer** as shown in Table 2.

2 points

Table 2: **Customer**

<i>CId</i>	<i>Name</i>	<i>Contact</i>
S001	David	11111
S002	Warner	12222, 12233
S003	Mathew	23221
S003	Mathew	23223
S005	Hayden	22111, 22333

- ☐ The relation **Customer** is in 2NF
- ☐ The relation **Customer** is in 1NF
- ☐ The relation **Customer** is not in 1NF
- ☐ The relation **Customer** has multivalued attributes.

Consider the relation **Customer** as shown in Table 3, having functional dependencies
 $F = \{ CId, Product \} \rightarrow Name,$
 $CId \rightarrow Name$

2 points

Table 3: **Customer**

<i>CId</i>	<i>Name</i>	<i>Product</i>
S001	David	Mobile
S002	Warner	Charger
S002	Warner	Laptop
S003	Mathew	Laptop
S003	Mathew	TV
S005	Hayden	TV

- ☐ The relation **Customer** is in 2NF
- ☐ The relation **Customer** is in 1NF
- ☐ The relation **Customer** is not in 2NF
- ☐ The relation **Customer** has multivalued attributes.

Which of the following is/are true regarding transitive dependency?

2 points

- ☐ A transitive dependency can occur only in a relation that has three or more attributes.
- ☐ A transitive dependency can occur only when there is partial dependency in a relation.
- ☐ Transitive dependency occurs when a non-key attribute determines another non-key attribute.
- ☐ All the above